

Electrical → Past Question

→ 2080/07/06

Date:

1 marks

→ 1 neg.

	TV
LSV	FV

1) Third angle projection → Figure quadran

2) (A/F, 7%, 8) → Notation

↳ Calculation of A. others given

3. Field on stator → AC excitation

4. Which donot take time value of money?

a) NPV b) IRR c) B/C d) Accounted rate of return

5. Unit of reactive power → KVAR

6. Space-height ration in indoor lighting
→ 1:2

7. Unit Fuse rating → Ampere

8. When secondary of which is not opened when primary energized → CT

9. Time constant of RC circuit → RC

10. When time constant increase, System is → Slower.

11. Starting torque of single phase motor
→ 0

12. Angle between main & Auxiliar winding in ϕ Induction motor.
→ 30 electrical [2] marks.

13. $f = 50 \text{ Hz}$, $V_{rms} = 30 \text{ V}$

eqⁿ of sine wave → $41.42 \sin 314t$

14. Constant load torque

→ Crane, Hoist etc.

15. Which do not provide energy? → Passive

16. In microwave heating, there is no
→ induction coils.

17. For general engineer → minimum qualification
→ B.E.

18. For professional " , " , "
→ Master

19. Which component is not used in Substation?
→ Exciter

20. The time to open the circuit breaker contact → Opening time

21. for 200 km → surge impedance = 400Ω
for 400 km → surge impedance = ?
→ 400Ω

22. for lossless line, P & Q directly proportional to
→ $\sin \delta$ & voltage drop respectively

23. In load bus → P & Q

23. Secondary energy sources are
→ Wood, animal waste, agricultural waste

24. Cheapest O.L.M cost → hydro

25. Thermal efficiency of diesel & petrol
→ Diesel has more

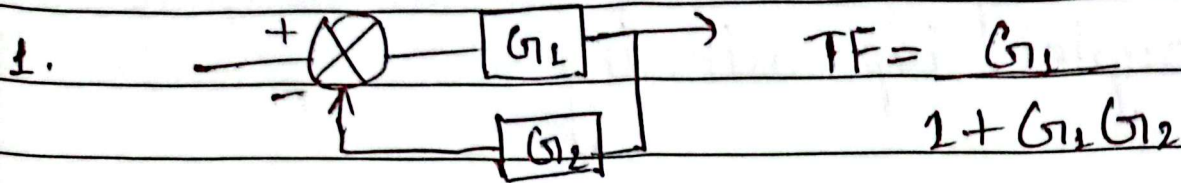
26. Power factor correction
→ to improve lagging to unity PF

27. Pumped hydro $\rightarrow$ Load factor & plant factor high.
28. Utility point of view for same energy consumption $\rightarrow$ high load factor.
29. Low Q & High P means $\rightarrow$ Higher PF
30. for lower cost per unit generation $\rightarrow$ Both LF & diversity factor should be high
31. In strain gauge $\rightarrow$ compressive stress $\rightarrow$ decreases resistance
32. Weighted Resistance method $\rightarrow$ SPST is used for connection in reference voltage or ground.
33. The blackstart unit is $\rightarrow$ Diesel power plant.
34. In open loop system wrong statement is $\rightarrow$ Non-linear inputs cause malfunctioned
35. In MOSFET $\rightarrow$ Drain, Source & Gate
36. Stranding increases $\rightarrow$ flexibility
37. Effective resistance of line increases by $\rightarrow$ skin, proximity effect & corona
38. Microprocessor based interfacing is required for $\rightarrow$ Memory & I/O
39. In power factor meter, phase diffⁿ between current in two coil?
 $\rightarrow$

40. Maximum emf in dc generator
→ Maximum flux linkage with conductor
41. Armature winding in negative slip conditions in synchronous motor
→ decreases the speed
42. Power angle equation, δ is
→ angle between electrical angle of rotor
43. 'Pantograph' is used for
→ Current collectors
44. Which method is used for power factor correction? → Synchronous Condenser
45. Modern HVAC transmission has
→ 12 pulse converter
46. If no. of voltage level is increased in inverter, THD level → decreases
47. In maxwell's ^{bridge} measurement of inductance
→ $1 < Q < 10$
48. TAP in transformer is used for
→ External protection.
49. Conservator tank is used for
→ Expansion & contraction of oil.
50. The flux ^{density} lags behind flux density magnetizing force (B behind H)
→ Hysteresis.

51. Lightning isolator protects against
→ indirect overvoltage due to indirect stroke
52. Earthing resistance in large power plant is → 0.5Ω .
53. ~~Severity in~~
53. According to stability, which fault is severe? → 3 phase fault
54. Major event in chart → Milestone Chart
55. In underground system, fault is detected using → Partial discharge
56. In rural distribution most economical is
→ 1 phase & wire
57. Fuse is never inserted in neutral.
58. Imbalance in rotor of induction motor is due to → non-uniform air gap
59. Increase in voltage cannot assured
→ Reduction in Corona loss
60. How many phase sequence in 3 phase AC system.
→ 2

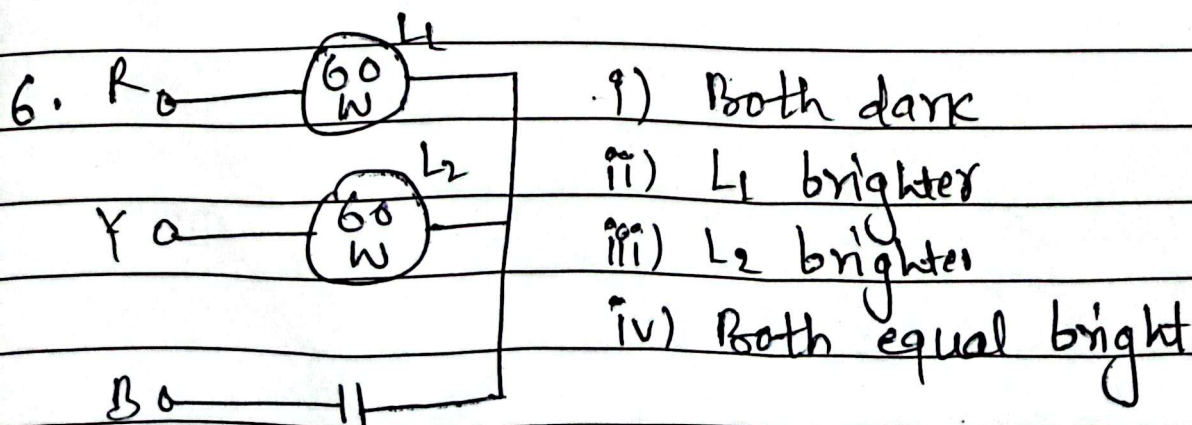
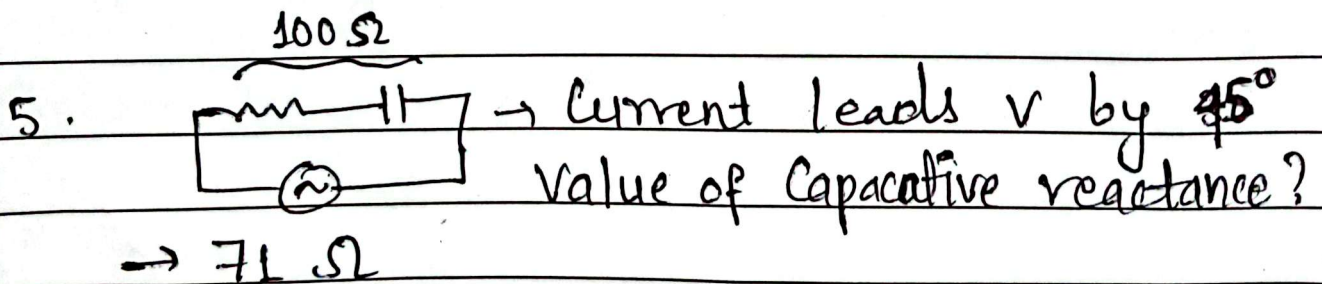
2 marks questions



2. PM & GM in bode plot
→ both positive

3. discharge vs time curve in hydropower station.
→ Hydrograph

4. In $\angle\phi$ Induction motor, main & auxiliary winding phase difference
→ 30° Electrical

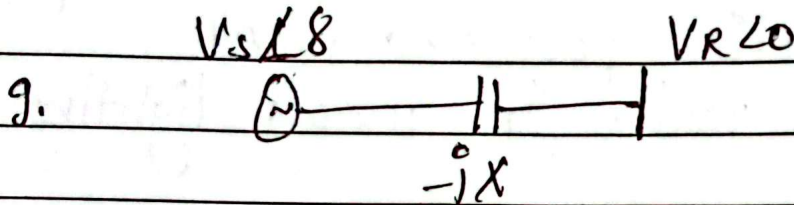


7. ~~Actual~~ Planned $\rightarrow 6000$, Actual $\rightarrow 7000$
 Earned $\rightarrow 8000$

Ans $\rightarrow$

8. Old: 0.2 pu , 11 kV , 500 MVA
 New: ? , 22 kV , 100 MVA

Ans $\rightarrow 0.01$



δ varies from $\rightarrow 0$ to 90°

10. We have to measure 15 V but our instrument can measure only upto 5 V .
 What to do? $\rightarrow$ Use of potentiometer

11. Blackstart is used to

$\rightarrow$ provide excitation to alternator during starting.

12. Characteristics of MOSFET & BJT

$\rightarrow$ MOSFET has high input impedance
 & BJT has lower

$\rightarrow$ MOSFET has lower switching loss
 & BJT has higher.

13. Stranding of fuse element Cause fusing current to
→ Decrease
14. Microprocessor based instrumentation timing.
15. When Contractor doesn't finish on time, what is compensation to Client?
→ Liquidated damage
16. Which is not used to maintain/improve voltage regulation
17. Price for 30 units (5A distribution)
→ Ans: A. 175
18. When a given transmission line is modeled in short transmission & then for same length & power, the system is modeled in medium transmission line, the voltage regulation will
→ increase

19. No. of parallel path in 2 poles duplex lap winding
→ 4

20. In Regenerative braking, energy supplied back to source does not depend on?

- a) Dead weight
- b) Track gradient
- c) Traction efficiency
- d) Starting & ending speed while braking

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